

# DataLab Data Dictionary

## Variable Reference Guide

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### Overview

This page provides a reference guide for all variables collected in the DataLab survey. Use this when analysing your data or preparing your research reports.

#### Quick Reference

The data dictionary is organised by station. Each variable includes its type, description, and expected range.

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### Participant Identifiers

Demographics

| Variable       | Type        | Description               | Values                                                      |
|----------------|-------------|---------------------------|-------------------------------------------------------------|
| participant_id | Integer     | Unique participant number | 1-60                                                        |
| lab_room       | Categorical | Lab assignment            | 304A, 304B                                                  |
| group_colour   | Categorical | Rotation group            | Red, Blue, Green, Yellow, Orange, Purple, Pink, White, Solo |
| session_date   | Date        | Auto-captured timestamp   | ISO format                                                  |

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### Sleep & Lifestyle

Baseline

| Variable        | Type       | Description                   | Range                         |
|-----------------|------------|-------------------------------|-------------------------------|
| sleep_hours     | Continuous | Hours of sleep last night     | 0-14                          |
| sleep_quality   | Ordinal    | Sleep quality rating          | 1 (Very poor) - 5 (Excellent) |
| caffeine_today  | Count      | Cups of tea/coffee today      | 0-10                          |
| exercise_week   | Count      | Days exercised this week      | 0-7                           |
| screen_estimate | Continuous | Daily phone screen time (hrs) | 0-16                          |

## Mood (Pre-Post Design)

### Pre-Session Baseline

| Variable   | Type    | Description     | Range |
|------------|---------|-----------------|-------|
| mood_pre   | Ordinal | Baseline mood   | 1-10  |
| energy_pre | Ordinal | Baseline energy | 1-10  |
| stress_pre | Ordinal | Baseline stress | 1-10  |

### Post-Session After Stations

| Variable    | Type    | Description         | Range |
|-------------|---------|---------------------|-------|
| mood_post   | Ordinal | Post-session mood   | 1-10  |
| energy_post | Ordinal | Post-session energy | 1-10  |
| stress_post | Ordinal | Post-session stress | 1-10  |

### Change Scores Calculated

| Variable      | Formula                  | Interpretation           |
|---------------|--------------------------|--------------------------|
| mood_change   | mood_post - mood_pre     | 1. <b>improved</b>       |
| energy_change | energy_post - energy_pre | 1. <b>more energetic</b> |
| stress_change | stress_post - stress_pre | • <b>less stressed</b>   |

## Personality (TIPI)

Big Five

The Ten-Item Personality Inventory measures the Big Five personality traits.

| Variable            | Type       | Description               | Range |
|---------------------|------------|---------------------------|-------|
| extraversion        | Continuous | Extraversion score        | 1-7   |
| agreeableness       | Continuous | Agreeableness score       | 1-7   |
| conscientiousness   | Continuous | Conscientiousness score   | 1-7   |
| emotional_stability | Continuous | Emotional stability score | 1-7   |
| openness            | Continuous | Openness to experience    | 1-7   |

## Station 1: Reflex Test

Reaction Time


**REFLEX TEST**  
*How fast is your nervous system?*

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**INSTRUCTIONS**

1. **Partner A** holds ruler at 30cm mark, hanging vertically
2. **Partner B** positions thumb and finger at 0cm — don't touch the ruler!
3. A drops ruler **without warning**
4. B catches it **as fast as possible**
5. Read the cm mark at the **top of your thumb**
6. Complete **5 trials**, then swap roles

**CONVERSION CHART (CM → MILLISECONDS)**

| CM | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10  |
|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| MS | 45  | 64  | 78  | 90  | 101 | 111 | 120 | 128 | 135 | 143 |
| CM | 11  | 12  | 13  | 14  | 15  | 16  | 17  | 18  | 19  | 20  |
| MS | 150 | 156 | 163 | 169 | 175 | 181 | 186 | 192 | 197 | 202 |
| CM | 21  | 22  | 23  | 24  | 25  | 26  | 27  | 28  | 29  | 30  |
| MS | 287 | 212 | 217 | 221 | 226 | 230 | 235 | 239 | 243 | 247 |

Formula:  $RT (ms) = \sqrt{(2d + 9.8t)} \times 1000$  where d = distance in metres

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**RECORD YOUR CATCHES (CM)**

|                      |                      |                      |                      |                      |
|----------------------|----------------------|----------------------|----------------------|----------------------|
| Trial 1              | Trial 2              | Trial 3              | Trial 4              | Trial 5              |
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |

Which hand did you use? ☐ Dominant ☐ Non-dominant

| Variable     | Type         | Description              | Range                     |
|--------------|--------------|--------------------------|---------------------------|
| rt_hand      | Categorical  | Hand used                | Dominant,<br>Non-dominant |
| rt_trial1_cm | - Continuous | Catch distance per trial | 0-30 cm                   |
| rt_trial5_cm |              |                          |                           |
| rt_missed    | Multi-select | Which trials missed      | Trial 1-5, No misses      |

## Calculated Variables

| Variable                       | Formula                     | Description        |
|--------------------------------|-----------------------------|--------------------|
| rt_trial1_ms<br>- rt_trial5_ms | $\sqrt{(2d/g) \times 1000}$ | RT in milliseconds |
| rt_mean_ms                     | Average of trials           | Mean reaction time |
| rt_best_ms                     | Minimum of trials           | Fastest reaction   |
| rt_improvement                 | Trial 1 - Trial 5           | Practice effect    |

### The Physics

Reaction time (ms) =  $\sqrt{(2 \times \text{distance\_cm} / 100 \div 9.81) \times 1000}$

A catch at 15cm  $\approx$  175ms reaction time

## Station 2: Mind Palace

Memory


**MIND PALACE**  
*Memory test materials*

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**STEP 1: DISTRACTOR TASK (DO THIS FIRST, FOR 60 SECONDS)**

Complete as many as you can. This prevents rehearsal of the word list.

|           |           |          |          |           |
|-----------|-----------|----------|----------|-----------|
| 47 + 36 = | 83 - 29 = | 6 × 7 =  | 54 ÷ 9 = | 28 + 45 = |
| 54 + 18 = | 91 - 47 = | 8 × 9 =  | 72 ÷ 8 = | 63 + 19 = |
| 26 + 57 = | 72 - 38 = | 4 × 12 = | 45 ÷ 5 = | 37 + 26 = |
| 63 + 29 = | 85 - 56 = | 7 × 8 =  | 63 ÷ 7 = | 52 + 31 = |
| 38 + 45 = | 64 - 27 = | 9 × 6 =  | 48 ÷ 6 = | 44 + 29 = |
| 52 + 39 = | 93 - 48 = | 5 × 11 = | 81 ÷ 9 = | 67 + 18 = |

**STEP 2: WORD RECALL (WRITE ALL THE WORDS YOU REMEMBER)**

|   |   |   |   |    |
|---|---|---|---|----|
| 1 | 2 | 3 | 4 | 5  |
| 6 | 7 | 8 | 9 | 10 |

Write words in any order – the numbers are just to help you count.


**TOTAL WORDS RECALLED:** ----- / 10

| Variable             | Type        | Description              | Range                |
|----------------------|-------------|--------------------------|----------------------|
| mem_total            | Count       | Words correctly recalled | 0-10                 |
| mem_pos1 - mem_pos10 | Categorical | Recalled each position?  | Yes, No, Un-<br>sure |

| Variable       | Type  | Description                  | Range     |
|----------------|-------|------------------------------|-----------|
| mem_intrusions | Count | False memories (wrong words) | 0-10      |
| mem_strategy   | Text  | Strategy used                | Open text |

### 💡 Serial Position Effect

Positions 1-3 (primacy) and 8-10 (recency) are typically recalled better than positions 4-7 (middle).

## Station 3: Time Warp

Estimation


**TIME WARP**  
Your brain's internal clock

**TASK A: TIME ESTIMATION**

1. Close your eyes or look down at the desk
2. When you hear "Start", begin sensing time passing
3. Say "Stop" when you believe exactly 60 seconds have passed
4. Your partner will record your actual time

**TASK B: LINE LENGTH**

1. Look at Line A (separate sheet)
2. Estimate length in cm
3. Repeat for Line B

*Don't measure!*

**TASK C: STRING LENGTH**

1. Feel String 1 (loop) without looking
2. Estimate length in cm
3. Repeat for String 2 (knotted)

*Touch only!*

**TASK D: TEMPERATURE**

1. Without checking any devices
2. Estimate room temperature
3. Record in °C

**RECORD YOUR ESTIMATES**

|           |          |          |           |           |          |
|-----------|----------|----------|-----------|-----------|----------|
| Time:     | Line A:  | Line B:  | String 1: | String 2: | Temp:    |
| ----- sec | ----- cm | ----- cm | ----- cm  | ----- cm  | ----- °C |

| Variable            | Type       | Description                 | Range      |
|---------------------|------------|-----------------------------|------------|
| time_estimate_sec   | Continuous | 60-second estimation        | ~30-90 sec |
| line_a_estimate_cm  | Continuous | Line A length estimate      | 1-25 cm    |
| line_b_estimate_cm  | Continuous | Line B length estimate      | 1-25 cm    |
| string1_estimate_cm | Continuous | String 1 estimate (tactile) | 1-40 cm    |
| string2_estimate_cm | Continuous | String 2 estimate (tactile) | 1-40 cm    |
| temp_estimate_c     | Continuous | Room temperature guess      | 15-30 °C   |

| Variable        | Type        | Description            | Range                                   |
|-----------------|-------------|------------------------|-----------------------------------------|
| time_counting   | Categorical | Counted during timing? | Yes steadily,<br>Yes some-<br>times, No |
| music_condition | Categorical | Lab condition          | silence, music                          |

## Calculated Variables

| Variable      | Formula           | Description            |
|---------------|-------------------|------------------------|
| time_error    | estimate - 60     | 1. <b>overestimate</b> |
| line_a_error  | estimate - actual | Estimation accuracy    |
| line_b_error  | estimate - actual | Estimation accuracy    |
| string1_error | estimate - actual | Tactile accuracy       |
| string2_error | estimate - actual | Tactile accuracy       |
| temp_error    | estimate - actual | Temperature accuracy   |

## Station 4: Sixth Sense

### Detection


**SIXTH SENSE**  
*Can you feel when someone is watching?*

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**INSTRUCTIONS**

- TARGET** sits with back to partner, eyes closed
- STARER** flips a coin: **Heads** = stare at back of partner's head; **Tails** = look away
- Hold for **15 seconds**, then tap partner's shoulder
- Target says **"Staring"** or **"Not staring"** and rates confidence
- Starer reveals the answer; record if correct
- Complete **6 trials** as target, then swap roles

**RECORDING SHEET**

| Trial | Actual                                                        | Your Guess                                                    | Confidence | Correct?                                              |
|-------|---------------------------------------------------------------|---------------------------------------------------------------|------------|-------------------------------------------------------|
| 1     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> X |
| 2     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> X |
| 3     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> X |
| 4     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> X |
| 5     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> X |
| 6     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> X |

Confidence: 1 = pure guess, 5 = absolutely certain


**TOTAL CORRECT:** \_\_\_\_\_ / 6

| Variable                           | Type          | Description          | Values                     |
|------------------------------------|---------------|----------------------|----------------------------|
| stare_t1_actual<br>stare_t6_actual | - Categorical | Was partner staring? | Staring, Not staring       |
| stare_t1_guess<br>stare_t6_guess   | - Categorical | What you guessed     | Staring, Not staring       |
| stare_t1_conf<br>stare_t6_conf     | - Ordinal     | Confidence           | 1 (Guessing) - 5 (Certain) |
| stare_solo                         | Categorical   | Completed solo?      | Yes, No                    |

### Calculated Variables

| Variable            | Description                       | Expected    |
|---------------------|-----------------------------------|-------------|
| stare_correct_total | Correct detections (0-6)          | ~3 (chance) |
| stare_hit_rate      | P(say staring   actually staring) | ~0.5        |
| stare_fa_rate       | P(say staring   not staring)      | ~0.5        |

## Station 5: Lady Luck

Randomness


**LADY LUCK**  
*Can humans be truly random?*

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**TASK A: GENERATE 20 "RANDOM" NUMBERS (1-10)**

Say each number aloud as it comes to mind — *don't plan ahead!*

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|    |    |    |    |    |    |    |    |    |    |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|    |    |    |    |    |    |    |    |    |    |

**TASK B: ROLL THE DICE 10 TIMES**

|   |   |   |   |   |   |   |   |   |    |
|---|---|---|---|---|---|---|---|---|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|   |   |   |   |   |   |   |   |   |    |

How many 6s did you roll? ..... (By chance, you'd expect about 1-2)

**TASK C: FLIP THE COIN 10 TIMES**

Record H (heads) or T (tails):

|   |   |   |   |   |   |   |   |   |    |
|---|---|---|---|---|---|---|---|---|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|   |   |   |   |   |   |   |   |   |    |

Total Heads: ..... Longest run of same outcome: .....

## Random Number Generation

| Variable                  | Type    | Description                | Range |
|---------------------------|---------|----------------------------|-------|
| rand_num1<br>- rand_num20 | Integer | Generated “random” numbers | 1-10  |

## Dice Rolling

| Variable                 | Type        | Description         | Range           |
|--------------------------|-------------|---------------------|-----------------|
| dice_predict             | Categorical | Prediction about 6s | Yes, No, Unsure |
| dice_roll1 - dice_roll10 | Integer     | Dice outcomes       | 1-6             |

## Coin Flipping

| Variable                    | Type        | Description    | Range             |
|-----------------------------|-------------|----------------|-------------------|
| coin_flip1<br>- coin_flip10 | Categorical | Coin outcomes  | Heads, Tails      |
| coin_longest_run            | Count       | Longest streak | same-outcome 1-10 |

## Calculated Variables

| Variable     | Description          | Expected          |
|--------------|----------------------|-------------------|
| rand_mean    | Mean of 20 numbers   | 5.5 (true random) |
| rand_repeats | Adjacent repetitions | ~2 (true random)  |
| dice_sixes   | Count of 6s rolled   | ~1.67             |
| coin_heads   | Count of heads       | ~5                |

## Station 6: Frontal Lob

Metacognition





DATALAB  
**FRONTAL LOB**  
Precision meets confidence

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**INSTRUCTIONS**

- Before throwing:** Rate your ball-throwing ability (1 = terrible, 10 = excellent)
- Stand at the marked line
- Throw ping pong balls at the target
- Complete **10 throws** with your dominant hand
- Count how many hit the target
- Optional:* Try 5 throws with your non-dominant hand

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**RECORD YOUR DATA**

|                            |                                   |                                 |                                                                                         |
|----------------------------|-----------------------------------|---------------------------------|-----------------------------------------------------------------------------------------|
| Self-rating:<br>----- / 10 | Dominant hand hits:<br>----- / 10 | Non-dominant hits:<br>----- / 5 | Which hand is dominant?<br><input type="checkbox"/> Left <input type="checkbox"/> Right |
|----------------------------|-----------------------------------|---------------------------------|-----------------------------------------------------------------------------------------|

| Variable             | Type        | Description                 | Range       |
|----------------------|-------------|-----------------------------|-------------|
| throw_self_rating    | Ordinal     | Self-rated ability (before) | 1-10        |
| throw_dom_success    | Count       | Dominant hand successes     | 0-10        |
| throw_nondom_success | Count       | Non-dominant successes      | 0-5         |
| throw_dom_hand       | Categorical | Dominant hand               | Left, Right |

## Calculated Variables

| Variable           | Formula                               | Interpretation                                      |
|--------------------|---------------------------------------|-----------------------------------------------------|
| throw_accuracy_pct | $\text{successes} \div 10 \times 100$ | Performance %                                       |
| throw_calibration  | accuracy - rating                     | 1. <b>underconfident</b> , - = <b>overconfident</b> |

### ! Calibration Score

- **Positive:** You performed better than you predicted (underconfident)
- **Negative:** You predicted better than you performed (overconfident)
- **Zero:** Perfect calibration!

## Session Feedback

End of Session

| Variable          | Type        | Description       | Range     |
|-------------------|-------------|-------------------|-----------|
| enjoyed_overall   | Ordinal     | Overall enjoyment | 1-10      |
| fav_station       | Categorical | Favourite station | 1-6       |
| least_fav_station | Categorical | Least favourite   | 1-6       |
| comments          | Text        | Open feedback     | Free text |

## Analysis Ideas

### Pre-Post Comparisons (Paired t-tests)

- Did mood improve during DataLab?
- Did stress decrease?
- Did energy change?

### Correlations

- Extraversion × mood change
- Sleep hours × energy change
- Conscientiousness × calibration

### Group Comparisons (Independent t-tests)

- Lab 304A (music) vs Lab 304B (silence) on time estimation
- Visual vs tactile estimation accuracy

### Signal Detection

- Calculate  $d'$  (sensitivity) for stare detection
- Test hit rate against chance (0.5)